

Two Simulations for Mystery Tokens

Topic 8.1 · TODAY'S PROBLEM

Suppose a company claims 10,000 mystery tokens are 50% red, 30% teal, and 20% gold. An SRS of 80 has the shown counts.

Leo: “Mix and redraw these same 80 tokens without replacement; recompute χ^2 for 1,000 trials.”

Amara: “For each trial, generate 80 independent colors with probabilities 0.50, 0.30, and 0.20, then recompute χ^2 .” In Amara’s trials, 186 statistics are at least the observed value.

Token sample

Color	Claim	Observed
Red	0.50	32
Teal	0.30	30
Gold	0.20	18

FIRST TAKE (5 min, on your sheet)

Which simulation can reveal whether the sample’s color discrepancies are unexpected? State what varies in one trial.

- DOK 1** (a) Calculate the three expected counts under the company’s claimed color model.
- DOK 2** (b) Compute the χ^2 contributions and total, give df , estimate the upper tail from Amara’s trials, and check 10 percent and expected counts.
- DOK 3** (c) Choose the simulation that answers the chance-variation question and judge surprise at 5 percent. Compare what each trial fixes or varies, use the simulated tail, explain the rejected output’s false assurance, and avoid claiming the model is proved.

Video + follow-along: u8_lesson1_live.html (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

